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Paleomagnetic Results from Precambrian Sedimentary Rocks in the Western United States

S. K RUNCORN

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Paleomagnetic Results from Precambrian Sedimentary Rocks in the Western United States

Abstract: Further paleomagnetic results from Precambrian formations of the western United States are described. The Sioux and Barron quartzites from Minnesota and Wisconsin, the Uinta Mountain Group from Utah, and the Apache Group from southern Arizona give poles in the east-central equatorial Pacific similar to those of the Grand Canyon Series from northern Arizona and the Belt Series from Montana. There is evidence that the rate of polar wandering varies with geological time, being least in the interval between 600 million years (m.y.) and 1000 m.y. This suggests a quiescent period in orogenic activity as is also indicated by the distribution of ages of metamorphic and igneous rocks. A comparison of these pole positions and those for Keweenawan rocks possibly suggests relative motion within the continent of North America.

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INTRODUCTION

This paper presents a further development of the paleomagnetic survey of red beds in the western United States, previously reported by Runcorn (1956a) and by Collinson and Runcorn (1960). The absence in this region of complicated tectonic movements enables one to correct paleomagnetic measurements on even the oldest formations for geological tilt with confidence. Also, compared with many other regions, such as the Appalachians, there is little chance that the rocks have acquired a secondary component of magnetization after formation as the result of stresses during tectonic activity. Further, the presence of numerous unmetamorphosed Precambrian red beds in the western States make possible a sufficient sampling of paleomagnetic directions for any trend to emerge in this long interval of geological time. For each formation, the extensive exposures enable one to make a fair sampling throughout its entire vertical extent.

The collection of samples, the measurement of their direction of magnetization, and the statistical analysis of the results have been carried out by the methods described by Collinson and others (1957) and by Collinson and Runcorn (1960). The vector mean of the directions of magnetization obtained from a certain geological formation is, after correction for geological tilt, transformed into a pole position on the present globe. The pole defines the axis of a geocentric magnetic dipole which would give a field at the site having the same direction as the mean paleomagnetic one. As

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the poles determined in this way for each geological period differ significantly for each continent, and in general are displaced farther from the present geographical pole with increasing age, the author (1956a; 1956b) postulated polar wandering and continental drift as explanations.

Some authors have doubted these conclusions. Billings (1960) referred to the absurdity which would result from applying these arguments to the field at the present time (or any other instant). But the deviations of the field from an axial geocentric dipole change very rapidly compared with geological time, and investigation of the cause of the geomagnetic secular variation shows that Billings' criticism is without substance. The relative rotation of the core and mantle, which accounts for the westward drift of the geomagnetic field and the irregular changes in the length of the day, causes the mean geomagnetic field to be symmetrical about the earth's axis of rotation (Runcorn, 1959a), when averaged over periods which are long compared with the secular variation time scale. A geological period is 10^3 longer; it is therefore not difficult to insure that enough of a formation has been sampled to average out the geomagnetic secular variation in the mean paleomagnetic direction.

However, the possibility that the field has in remote geological time been an axial-multipole field cannot be rejected *a priori*. This explanation of the discrepancies between polar-wandering curves determined from the different continents has been tested and does not satisfy the data for Upper Paleozoic and Mesozoic rocks (Runcorn, 1959b). Blackett and others (1960) have tried to present the paleomagnetic evidence for continental drift without any assumptions about the long-term behavior of the geomagnetic field. The comparison of paleomagnetic results for one age from sites within one continent cannot yet be made with sufficient accuracy to test directly the dipole character of the mean geomagnetic field.

In response to our suggestion that a set of paleomagnetic observations merited an exact statistical treatment, Fisher (1953) gave a treatment for dispersion on a sphere, analogous to that of Gaussian statistics. Some applications of the method by the author have been criticized because they tend to overestimate the accuracy of the means (Cox and Doell, 1960). This criticism arises from the fact that there are two different views on how to compute the semiangle of the cone of confidence for a mean. The question is whether it is correct to use the number of discs measured or the number of rock samples as the number of independent measurements. If the number of rock samples is used, the cone of confidence is increased by the square root of the average number of discs taken per sample. In the absence of a real understanding of the time relationships within a series of deltaic sediments, it is not possible to be dogmatic about this point, and the cone of confidence must be regarded as a useful, although not infallible, guide to the accuracy of the mean. Of course, this cone is a measure of the possible error arising from the incompleteness of sampling of the formation and reveals nothing, by itself, concerning possible systematic errors.

The most satisfactory rebuttal of these criticisms of the bases of paleomagnetic arguments is shown in Figure 1. The polar-wandering curves for Britain and North America given by Collinson and Runcorn (1960) were drawn to pass smoothly through the poles determined by Creer and others (1957) from British rocks, and by Collinson and Runcorn (1960) from their paleomagnetic survey of North American rocks. These paths, the individual poles having been omitted, are shown in Figure 1. As many other paleomagnetic results from other laboratories are now available, it is desirable to examine to what extent these support the paths of Figure 1. The numbers of specimens used, the methods of measurement and analysis, and the complexity of the geological environment are all different. Thus it is hard to justify each pole being given equal weight in calculating the mean pole for each geological period. Any more elaborate method also raises objections, so that the simplest one is used here. Mean poles for the different periods, computed from Irving's (1960–1961) tables, are shown in Figure 1 and are connected by arrows to the part of the polar-wandering paths to which they correspond. The agreement is good except for the Jurassic pole determined for Europe and the Precambrian pole determined for North America. Creer and others (1960) and Runcorn (1962) showed that the paleomagnetic results on the European Jurassic rocks are so wildly scattered that one must conclude that the lithologies available for this period are not suitable for paleomagnetic work, probably because the magnetizations are too weak. The discrepancy in the Precambrian pole for North

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America might be expected because of the difficulty of age correlation, but for the reasons given heretofore, the study of further observations is worthwhile. This paper is concerned, therefore, with the results of a further survey of the Precambrian of the western United States.

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I have had much valuable help in the field from Mr. David G. Smith (on the Precambrian rocks from the Middle West), Mr. Andrew F. Bateman (Montana), Mr. G. E. Untermann and Dr. M. D. Crittenden (Utah), Dr. J. F. Lance, Mr. B. Babbitt, and Mr. F. Pashley (Arizona), and the University of Utah Geology Department, to all of whom my sincere thanks are given.

CONVENTIONS USED

Points on the lower hemisphere of the stereographic projection, *i.e.* representing directions with positive (downward) angles of inclination, are represented by solid circles. Points on the upper hemispheres of the projection, *i.e.* representing directions with negative (upward) angles of inclination, are represented by open circles. Directions from discs cut from the same core are connected together by a full line, and directions from cores from the same rock sample are connected by a broken line. The samples are numbered from the bottom of the section upward; missing numbers represent samples either too weakly

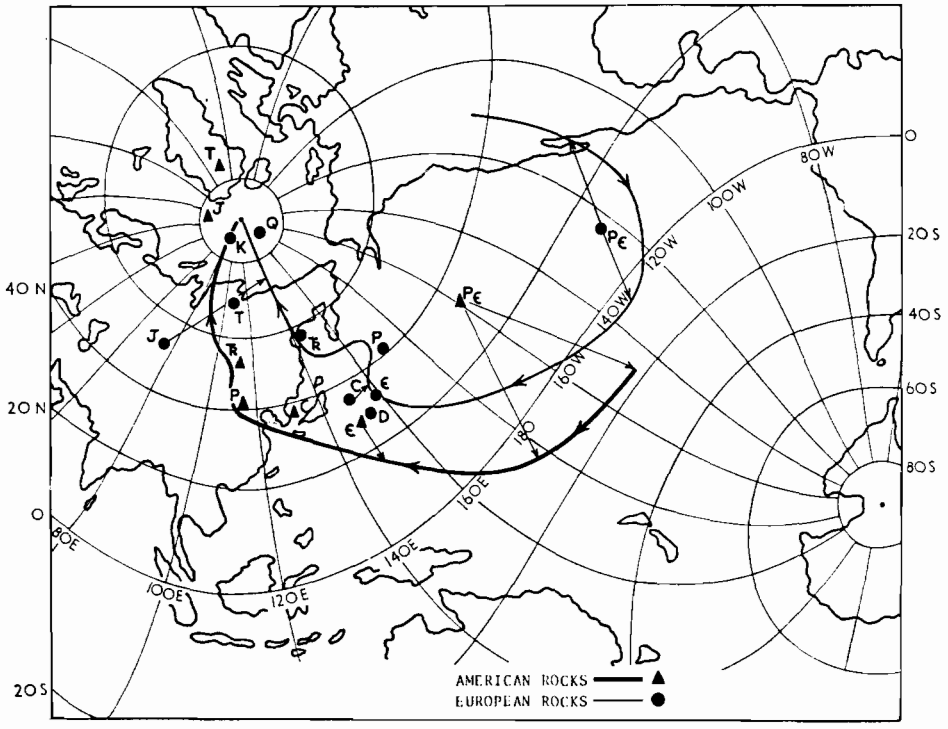


Figure 1. Polar wandering curves for North America given by Collinson and Runcorn (1960), and for Europe, by Collinson and others (1957) compared with mean poles computed from Irving's (1960–1961) Tables I–III. Oblique Mercator projection: angular distance scale uniform in central half of map

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magnetized to be measured or too fragile to be satisfactorily cut to shape.

The direction which the geomagnetic field would have at the site if the earth's field were that of a dipole orientated along the present geographical axis is plotted on each stereographic projection by a square (solid for sites in the present northern hemisphere). The mean paleomagnetic direction at a site is calculated both from all the discs and from the mean directions of each rock sample, and each is plotted by a star. Erratic results, although plotted, are not used in this calculation. The center of the star is solid if the direction has a positive angle of inclination and an open circle if the direction has a negative angle of inclination. Described around the mean point is the 95 per cent circle or cone of confidence of radius α. Samples from some formations in which the bedding planes were easily recognizable were cored so that the axes of the cores were perpendicular to these bedding planes. The declination and inclination of the magnetization of the discs cut from these cores were thus automatically corrected for tectonic dip. For all other samples the cores were drilled perpendicular to the oriented surfaces. The strike and dip of the latter were then used to correct the measurements so that the directions refer to the position of the rocks today. A further correction using the local strike and dip of the beds was made to restore the direction of magnetization to its direction before the tectonic movement. The declination and inclination given in Table 1 are corrected for local geological dip.

PRECAMBRIAN PALEOMAGNETIC MEASUREMENTS

General Statement

Creer and others (1957) (from a study of European rocks) and Runcorn (1956a) (from a study of North American rocks) showed that the pole has moved around the North Pacific at a mean rate of one third of a degree per m.y. since the Precambrian. Although the polar-wandering curve for North America is displaced westward from that for Europe (Runcorn, 1956b; Collinson and Runcorn, 1960), observations from both continents suggested that the pole had been in North America in earlier Precambrian times.

Collinson and Runcorn (1960) described many measurements on the Belt and Grand Canyon Series in Montana and Arizona and showed that the poles are grouped closely in the equatorial mid-Pacific. Creer and others (1957) had shown that the Diabaig group of the late Precambrian Torridonian sandstone of northwest Scotland which underlies and is thought to be much older than the two upper groups of the Torridonian gave a pole in Arizona much farther back along the polar-wandering path around the North Pacific. Further, Graham (1953) had shown that the Michigan dikes of Precambrian age gave a pole in North America as well. It therefore seemed possible that during the latter part of the Precambrian, the pole might have passed from North America to the equatorial mid-Pacific. The paleomagnetic results on late Precambrian formations of the western United States might therefore enable them, even with the present limitations in accuracy, to be put in order of geological age. I now describe some new paleomagnetic results from Precambrian sedimentary rocks.

Barron and Sioux Quartzites

The Barron quartzite of Wisconsin is in most places a fine-grained, flesh-pink, well-bedded rock, almost wholly of quartz. Cross-bedding is rare, and folding and faulting are gentle, the dips being usually less than 12°. The Sioux Quartzite of Minnesota is thought to be equivalent. It crops out in the Minnesota valley in a series of terraces along the north wall, where it consists of red quartzite, sandstones, and conglomerates. Samples were taken in four localities, and the stereograms of the results are shown in Figure 2 (A-D).

The exposures at each of the localities are poor, and the specimens too few to produce symmetrically grouped stereograms. In spite of this, the mean directions at the different sites are in reasonable agreement. Thus one can conclude that the magnetization is predominantly due to a single component, which, being significantly different from the present dipole field, is likely to be an original magnetization.

Goldich and others (1961) date the Sioux Quartzite as at least 1 billion years, but think it is very much older; therefore the paleomagnetic results, though meager, are of considerable interest. The nearly vertical directions of magnetization imply a pole closer to North America than the Belt Series and Grand Canyon Series.

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Big Cottonwood Formation

Purple shales of the Big Cottonwood formation below the supposed tillite in the Wasatch Range were collected in Big Cottonwood Canyon. The magnetizations of these shales were found to be affected by an intrusion, and the original magnetization (Fig. 3) is found only in exposures of this formation remote from this igneous body.

Pioneer Shale

An important group of Precambrian sedimentary rocks, the Apache Group, described by Darton (1925), crop out in southern Arizona. The purple Pioneer Shale is the only member of this group that is strongly magnetized. Collections were made in three areas (Fig. 4A-C).

All the directions shown in Figure 4 are symmetrically grouped except at C, where there is a group of specimens (5, 14, 15, 18, 19, 20) which are very much more strongly magnetized than the others. Therefore it is thought that they have acquired a strong component of secondary magnetization and are thus excluded from the present statistical analysis.

Uinta Mountain Group

Forrester (1937) describes the late Precambrian Uinta Mountain Group which in most places is a fairly coarse distinctly bedded red rock. The directions of magnetization are shown in Figure 5.

Collinson and Runcorn (1960) made measurements on a group of red sandstones from the north flank of the Uinta Mountains. The sandstones lie below a prominent outcrop of the Madison limestone, at a road cutting in Hunssinger Creek, on the Vernal-Manila road, Utah. At that time the outcrop was thought to be the Lodore Formation of Cambrian age, but as the pole position for this formation grouped itself with those of the Belt and Grand Canyon Series, Collinson and Runcorn (1960) stated that a late Precambrian or at least a very early Cambrian age at this locality would be more appropriate. Recent field studies place it in the Uinta Mountain Group.

Belt Series

Further collections were made in the red sedimentary rocks of the Belt Series, recently described by Ross (1959). Stereograms of the results are shown in Figure 6 (A, B).

Grand Canyon Series

Samples were taken of the distinctly bedded Hakatai Shale and the Bass Limestone from the Kabaib trail on the south rim of the Grand Canyon, northern Arizona (Fig. 7A, B). The mean directions, pole positions, and statistical data for all these formations are given in Table 1.

DISCUSSION OF RESULTS

In any paleomagnetic survey, it is necessary to ask first whether there is *prima facie* evidence that the magnetizations were original and have remained unchanged in direction since they were acquired. A decrease in intensity of the magnetization without change in direction is bound to occur through thermal fluctuation in the domains of the magnetized iron oxide grains and is of no consequence for this work. A symmetrical grouping of directions, significantly different from the present dipole field, shows that the remanent magnetization has not been acquired in the last million years. A markedly asymmetrical grouping suggests the presence of primary and secondary magnetizations and precludes any conclusion being drawn concerning the paleofield, until the components can be separated by demagnetization experiments or by other means. In the former case if metamorphism has not occurred, it is legitimate to postulate that the original magnetizations were acquired at the time of deposition or within a geological period of this time. This may best be tested by the comparison of formations of similar age within one continent.

Figure 8 shows that these paleomagnetic results pass this test, to a first approximation, for the pole positions from the Belt Series, Grand Canyon Series, Uinta Mountain Group, and Apache Group form a closely grouped pattern in the equatorial mid-Pacific. There are of course geological reasons for supposing that the first three are contemporaneous. Whereas a single geological formation such as the Chugwater of Triassic age in Wyoming gives pole positions scattered over a circle of about 5-degree radius (*See* Collinson and Runcorn, 1960, Figure 27), the pole positions for these Precambrian rocks form a distinctly elongated distribution over 40°. This is consistent with the greater interval of time which they undoubtedly span.

In some of the distributions there is some

TABLE 1. DIRECTIONS OF MAGNETIZATION AND POLE POSITIONS FOR PRECAMBRIAN FORMATIONS

Fig. no.	Formation	Locality	Lat.	Long.	Thickness sampled (ft.)	Strike	Dip	S	N	N'	R	D	I	α	K	Pole positions		$d\psi$	$d\chi$
																Lat.	Long.		
7B	Bass Limestone (Grand Canyon Series)	Grand Canyon	36° N.	112° W.	30	..	..	8	..	..	..	N. 231° E.	+36°	2°	..	16° S.	162° W.	2°	1°
7A	Hakatai Shale (Grand Canyon Series)	Grand Canyon	36° N.	112° W.	30	N. 41° E.	12° E.	9	50	50	8.95	N. 224° E.	+74°	4°	151.9	12° N.	133° W.	7°	7°
6B	Missoula group (Belt Series)	Road cutting, 14 miles west of Essex, Mont.	48° N.	114° W.	400	N. 151° E.	57° NE.	39	143	143	31.33	N. 38° E.	−27°	6°	19.2	19° S.	153° W.	3°	6°
6A	Grinnell Argillite (Belt Series)	Bad Rock Canyon, Montana	48° N.	114° W.	..	N. 113° E.	24° NE.	9	48	48	8.32	N. 240° E.	+7°	16°	11.8	17° S.	179° W.	8°	16°
5	Uinta Mtn. Group	Road cutting on Manila-Vernal Rd., Utah	41° N.	110° W.	30	N. 72° E.	19° NW.	8	33	33	5.43	N. 53° E.	−7°	24°	8.8	24° S.	171° W.	12°	24°
4C	Pioneer Shale (Apache Group)	North end of Slate Mtn., Ariz.	33° N.	112° W.	..	N. 64° E.	23° NW.	21	74	49	9.58	N. 18° E.	−52°	23°	4.6	23° S.	128° W.	21°	31°

4B	Pioneer Shale (Apache Group)	Roosevelt Dam, Ariz.	34° N.	111° W.	85	N. 35° W.	27° NE.	13	49	49	10.74	N. 29° E.	−33°	20°	5.3	31° S.	144° W.	13°	23°
4A	Pioneer Shale (Apache Group)	Pinal Creek, Ariz.	33° N.	111° W.	56	N. 15° E.	26° E.	15	71	71	13.71	N. 18° E.	−48°	12°	10.9	25° S.	128° W.	10°	16°
3	Big Cottonwood formation	Big Cottonwood Canyon, Utah	41° N.	111° W.	100	N. 293° E.	47° N.	7	26	26	..	N. 78° E.	−31°	18°	..	2° N.	180° W.	19°	11°
2D	Barron quartzite	Silver Creek, Barron County, Wisc.	45° N.	92° W.	100	N. 90° E.	14° N.	8	35	35	6.81	N. 273° E.	+79°	25°	5.9	42° N.	122° W.	45°	47°
2C	Barron quartzite	County roads C and NN, Barron County, Wisc.	45° N.	92° W.	100	N. 40° E.	6° NW.	9	27	27	7.66	N. 208° E.	+62°	12°	20.7	2° N.	112° W.	15°	19°
2B	Sioux Quartzite	Bank of Minnesota River	44° N.	94° W.	30	N. 130° E.	22° NE.	8	34	34	4.24	N. 243° E.	+56°	58°	1.9	9° N.	141° W.	60°	83°
2A	Sioux Quartzite	Quarry 4 miles west of Courtland, Nicollet County, Minn.	44° N.	94° W.	40	N. 130° E.	22° NE.	8	35	35	6.12	N. 122° E.	+79°	25°	6.8	31° N.	73° W.	45°	47°

S—Number of samples collected; *N*—Number of discs measured; *N′*—Number of discs used in the determination of the mean direction; *R*—Vector sum of the *N′* directions (each assigned unit length); *D*—Mean declination; *I*—Mean inclination; α—Semi-angle of cone of confidence; *K*—Precision; $d\psi$, $d\chi$—Semiaxes of the oval of confidence around each pole position.

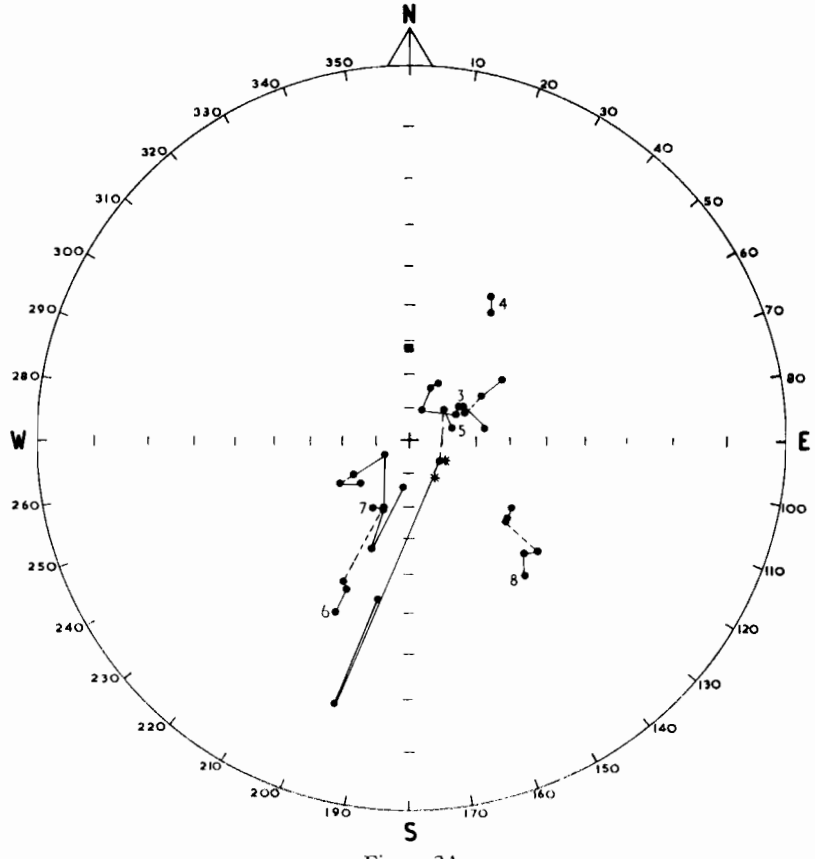


Figure 2A

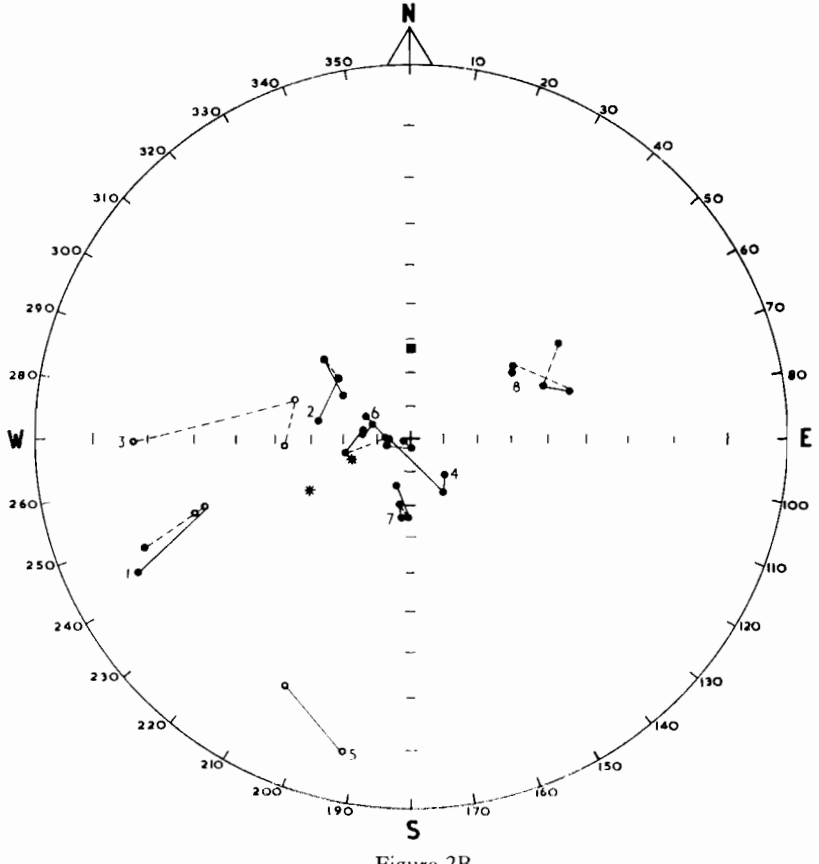


Figure 2B

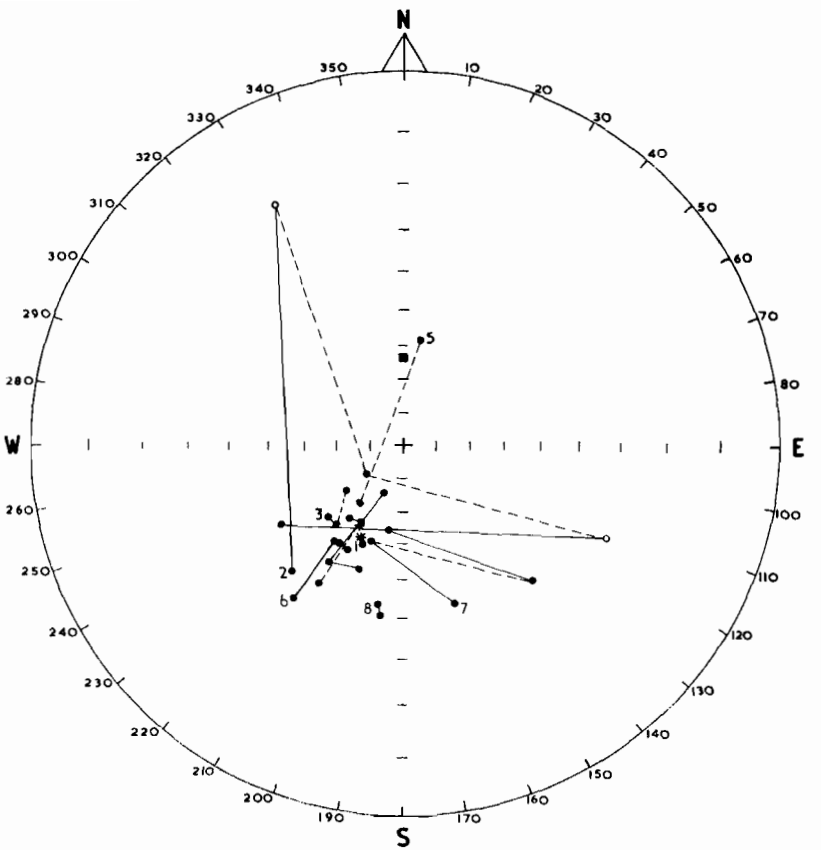


Figure 2C

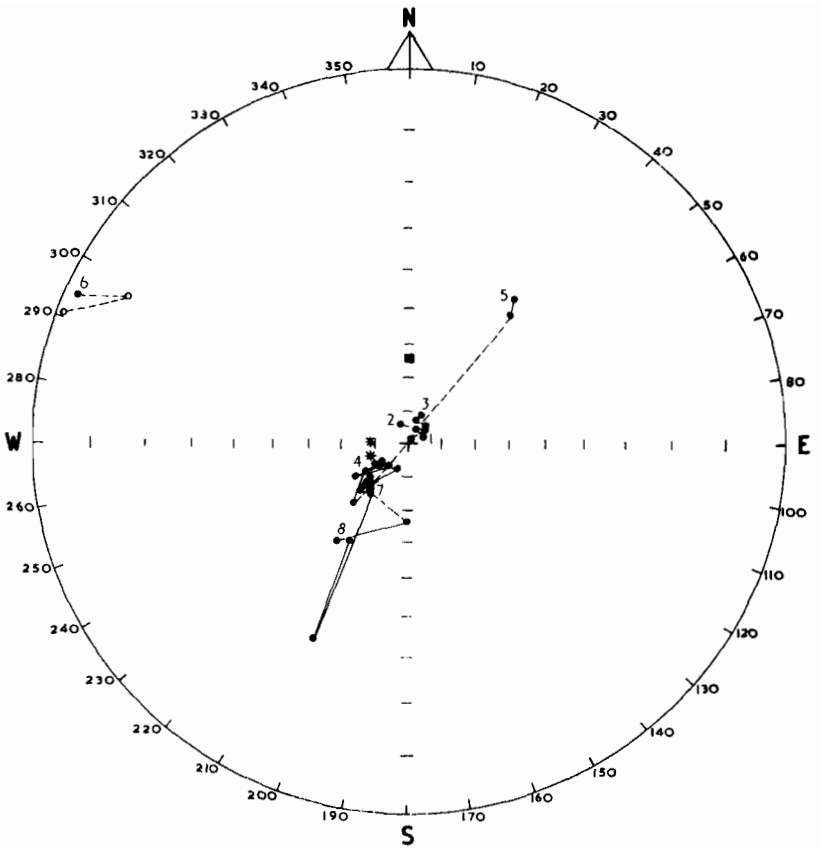


Figure 2D

Figure 2. Stereograms of the results of samples taken from the Sioux and Barron quartzites, Wisconsin and Minnesota. A, Sioux quartzite, quarry 4 miles west of Courtland, on highway 14, Nicollet County, Minn.; B, Sioux quartzite, one quarter of a mile southwest of locality (A) on the northeast bank of the Minnesota River; C, Barron quartzite, at intersection of County roads C and NN, Township 35 N., Barron County, Wisc.; D, Barron quartzite, Anderson's quarry, Silver Creek, Barron County, Wisc.

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suggestion of a secondary component of magnetization acquired in Quaternary time, which can usually be recognized because the directions lie along a great circle passing through the present dipole field. Runcorn (1956a) drew attention to the occurrence of this phenomenon

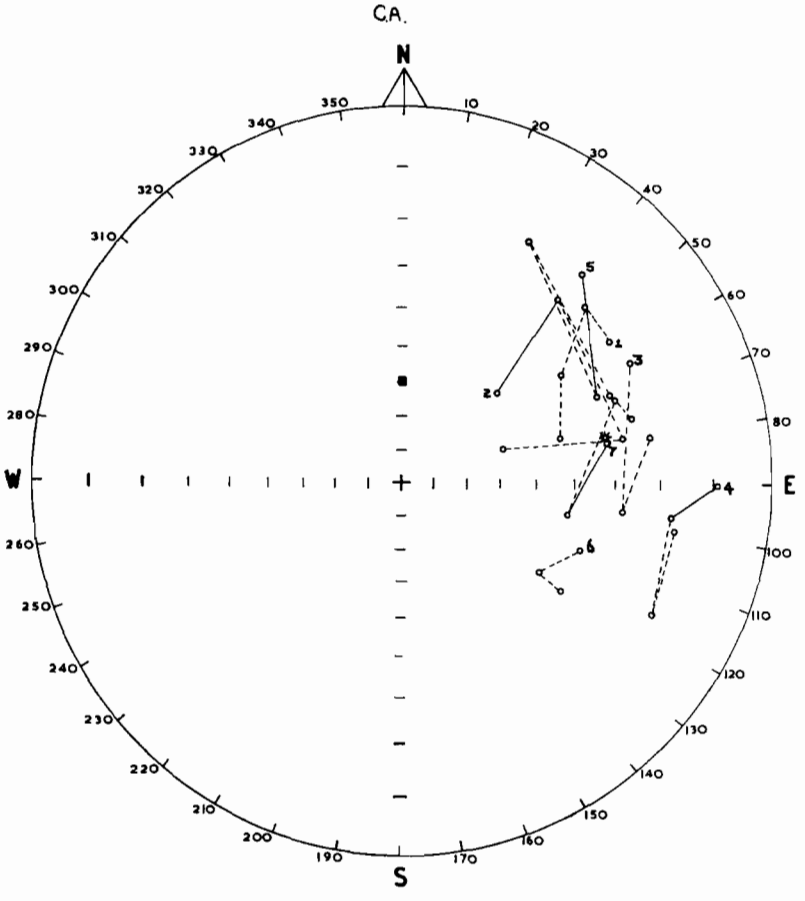


Figure 3. Stereogram of shales from Big Cottonwood formation, Moss ledge picnic ground, Big Cottonwood Canyon, Wasatch Range, Utah (See M. D. Crittenden, Jr., 1964, Geology of Mount Aire quadrangle, Salt Lake County, Utah: U.S. Geol. Survey Map GQ-379

in red beds from the western United States. If they are flat-lying, Runcorn (1956b) proved that the longitude, but not the latitude, of the computed pole position is still correct. Varying amounts of this component of secondary magnetization cannot explain the distribution of Precambrian poles. It may tentatively be concluded that the distribution is essentially due to age differences in the formations sampled.

From Figure 26 of Collinson and Runcorn (1960), one can see that displacements of 50° of the pole have occurred since the end of Paleozoic time and between the middle of the Early Paleozoic and the end of the Paleozoic, that is, displacements of about one fifth of a degree per million years, considerably less than that previously estimated by Creer and others (1957). If this rate were extrapolated, the date of the latest poles in the Precambrian belt described above would be 700 m.y., and the earliest would be 900 m.y. The pole for the Hakatai Shale from the Kaibab trail on the south rim of the Grand Canyon obtained by Runcorn (1956a) is definitely north and east of the main group, but the position is confirmed by measurements on different beds of the same outcrop reported in this paper. Yet the Shinumo Quartzite above it, the Bass Limestone below it, and the Hakatai Shale on the

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Kaibab trail give poles, determined by Collinson and Runcorn (1960) close together at the western end of the belt. The Bass Limestone in this paper gives a similar result. An explanation of this anomaly has not been found, although the beds in question are close to the Bright Angel fault. Apart from this anomaly, the Grand Canyon Series gives poles near the youngest of the Belt Series poles, and the Apache Group, near the earliest poles of the Belt Series. The poles of the Bonito Canyon quartzite as described by Collinson and Runcorn (1960) fall close to the poles of the Apache Group.

The question now arises as to the relationship of these results to the radioactive ages of these formations. Radioactive age determinations on diabase intrusions into the Apache Group have been obtained by Silver (1960) who dates it at greater than 1100–1200 m.y. Bearing in mind that the outcrops of the Sioux and Barron quartzites are poor, and the results scattered, paleomagnetism and radioactive age determinations agree in ascribing a similar age, greater than 1100 m.y., to the Apache Group and the Sioux and Barron quartzites. The paleomagnetic results would suggest that the Grand Canyon Series, the Uinta Mountain quartzite, and much of the Belt Series have a younger age, which could not be less than 800 m.y. The Belt Series is now dated at at least 1100 m.y. (Gulbrandsen and others, 1963). Dates, therefore, based on the assumption that the polar-wandering rate was uniform throughout the Precambrian, and equal to that since the Cambrian, seem low compared to the dates obtained by radioactive methods.

Thus the tentative conclusion is reached that the rate of polar wandering has not been constant through geological time, and that it was a minimum ($<1/10°$/m.y.) 500–1000 m.y. ago. Gold (1955) suggested that mountain building would produce polar displacements, and Runcorn (1957), that flow in the mantle would do so. These papers did not consider the possibility that continental displacements could occur, but these displacements do produce relative motion between a continent and the pole. If, however, the low rate of polar wandering in the interval 500–1000 m.y. is found to be a world-wide phenomenon, one must conclude that crustal movements had been at a minimum during this period.

In developing an explanation of continental drift, the writer (1962) drew attention to the significance of the discovery by Gastil (1960) and others of a tendency for the ages of metamorphic and igneous rocks to cluster about 2600 m.y., 1800 m.y., 1100 m.y., and 200–400 m.y. By whatever mechanism these world-wide events are caused, by changes in the convection pattern or otherwise, they represent periods of marked upheaval in the earth's crust, separated by comparatively quiescent intervals. There seems therefore to be some support from the paleomagnetic record for the reality of the late Precambrian gap in the radioactive age histogram.

The results discussed in this paper on the Belt Series agree very well with those determined by Black (1963) for the Purcell System of Canada, the mean pole of which is given as 10°S., 148°W. However, the work of DuBois (1962) on the Keweenawan rocks gives results which are considerably at variance with those described in this paper. Figure 9 shows the pole positions from Table 22 of DuBois' (1962) paper. The formations for which he was able to show magnetic stability are those grouped in higher latitudes than the pole positions given in this paper. It appears that either the Keweenawan rocks are of different age than those described in this paper, or that prior to the mountain building event of 1000 m.y. ago, the relative positions of the western United States and the region around Lake Superior were different. Our knowledge of the radioactive ages of these formations does not permit us to decide this issue. Eckelmann and Miller (1956) dated a pitchblende vein in the Coeur d'Alene region of Idaho (St. Regis quartzite) as being at least 1200 m.y. old, in agreement with the age of the Sullivan ore deposit at Kimberley, British Columbia, contained in a host rock of Purcell age (Wanless and Leech, 1958).

If these minimum ages for the time of Beltian deposition are the actual ages, they appear to be the same as that of the Keweenawan deposition for the Duluth Gabbro of middle Keweenawan age, dated by Goldich and others (1961) as 1120 m.y. It therefore appears that relative movements within one continent may be of importance in considering the interpretation of paleomagnetic results of the earlier Precambrian.

STABILITY OF MAGNETIZATION

Laboratory methods, *i.e.* thermal and alternating-current demagnetization, of eliminating secondary magnetization from the sedimentary rocks, and obtaining more exact measurements of the original magnetization are in progress. Because the intensity of magnetization of many

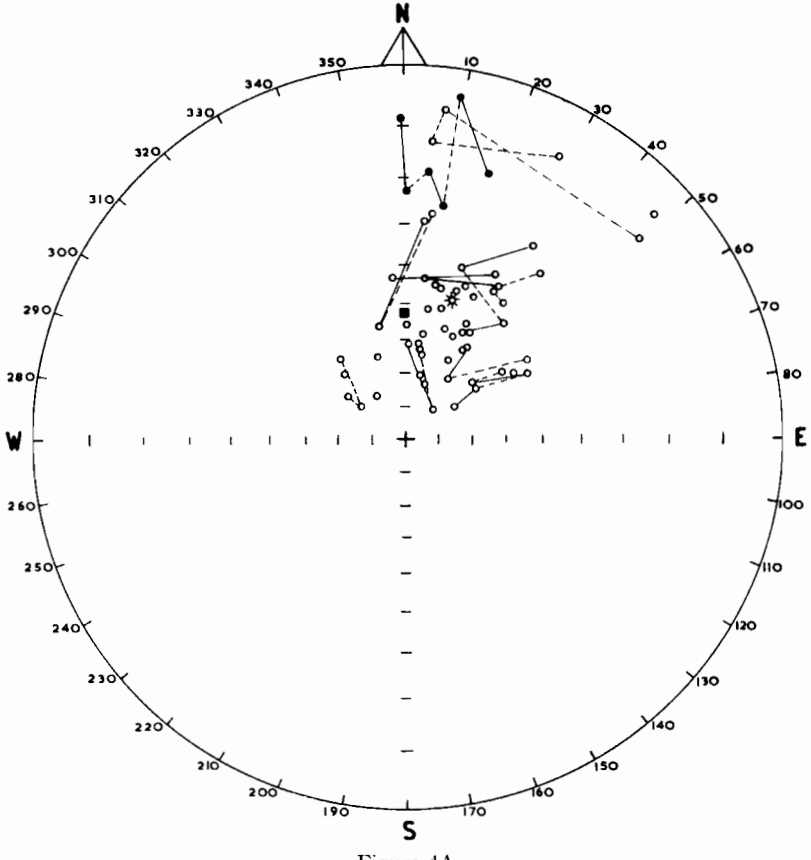


Figure 4A

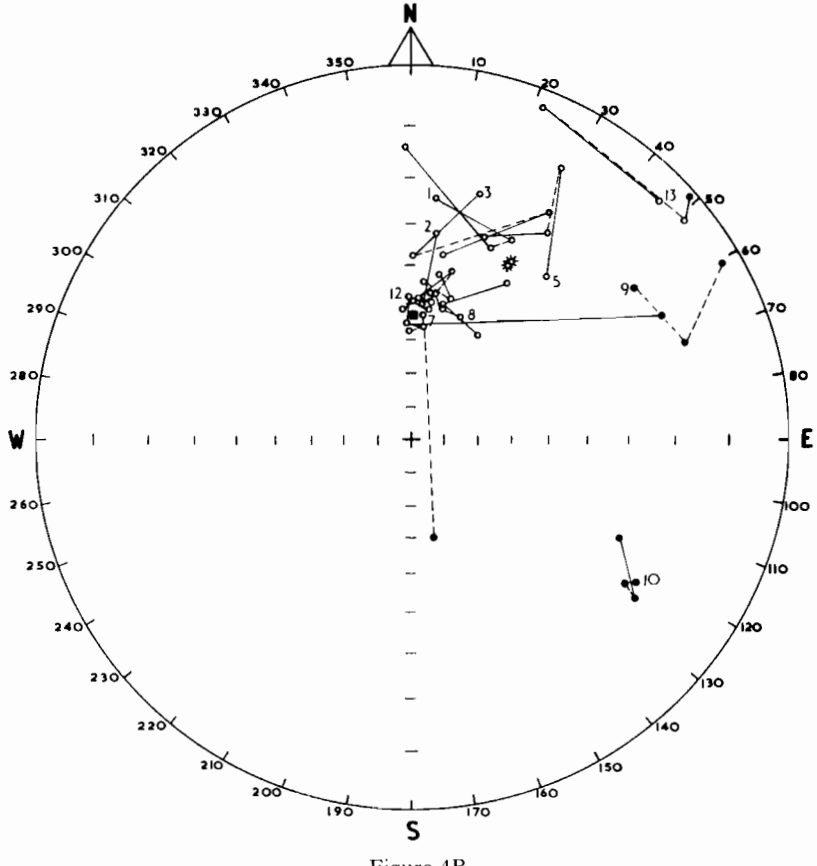


Figure 4B

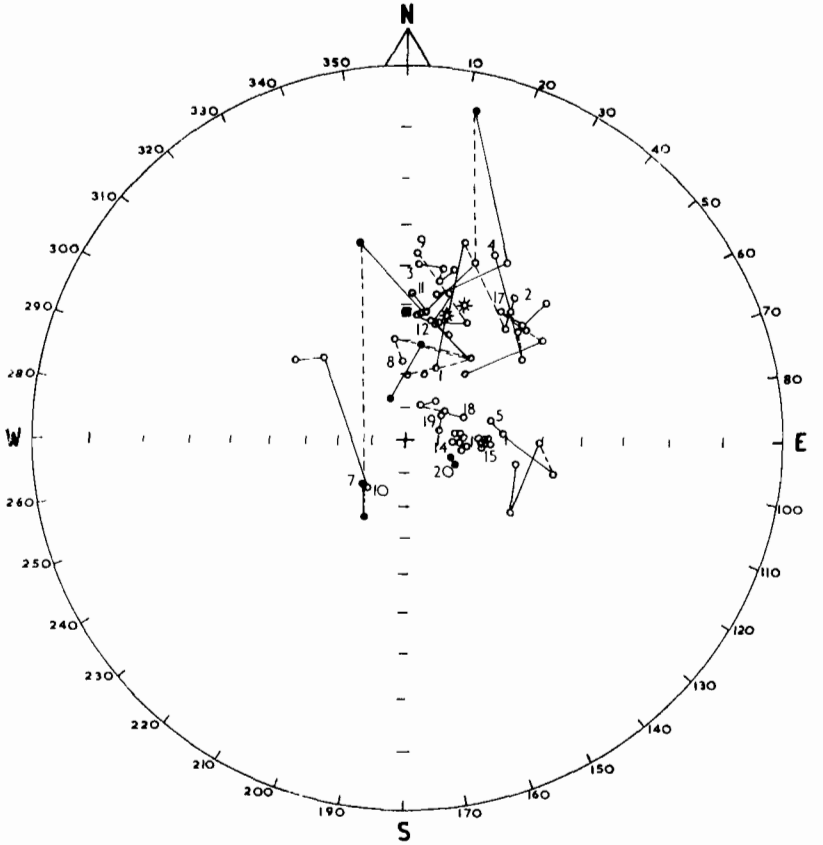


Figure 4C

Figure 4. Stereograms of Pioneer Shale, southern Arizona. A, Pinal Creek, near the Apache trail, southwest of Globe; B, Near Roosevelt Dam; C, North end of Slate Mountain

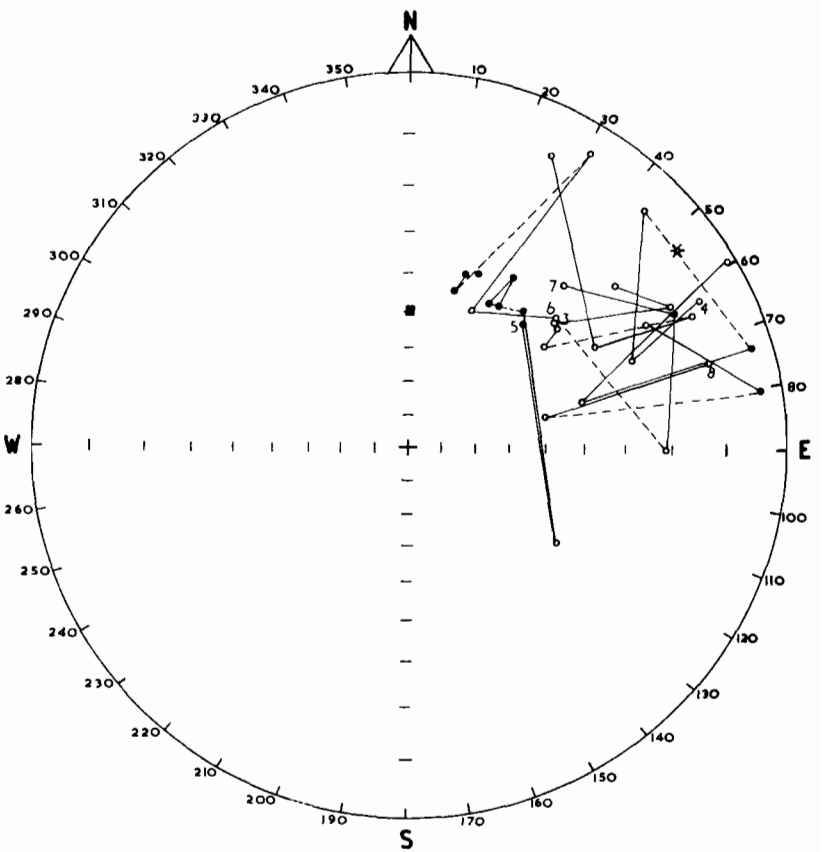


Figure 5

Figure 5. Stereogram of Uinta Mountain Group, Utah, as described by Forrester (1937)

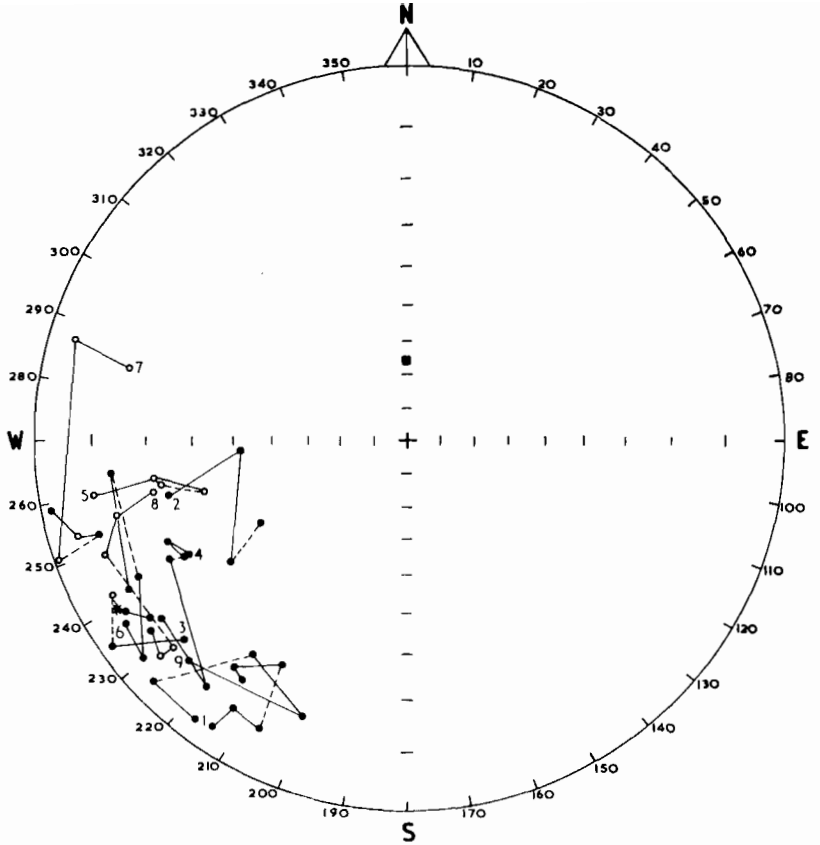


Figure 6A

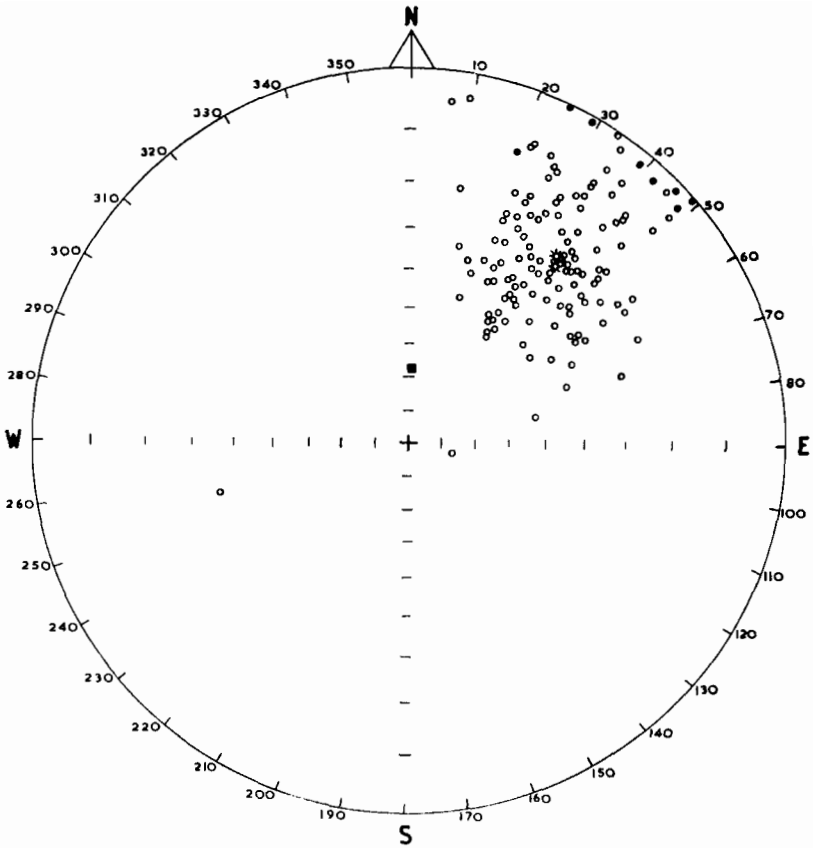


Figure 6B

Figure 6. Stereograms of samples from the sediments of the Belt Series, Montana. A, Grinnell Argillite, Bad Rock Canyon, along the Great Northern railway track; B, Missoula group, on the south side of the highway extending southeast of Glacier National Park, 14 miles west of Essex. This rock was formerly assigned to the Spokane Formation, the red band in the Siyeh, but has now been assigned by Ross (1963) to the Missoula group.

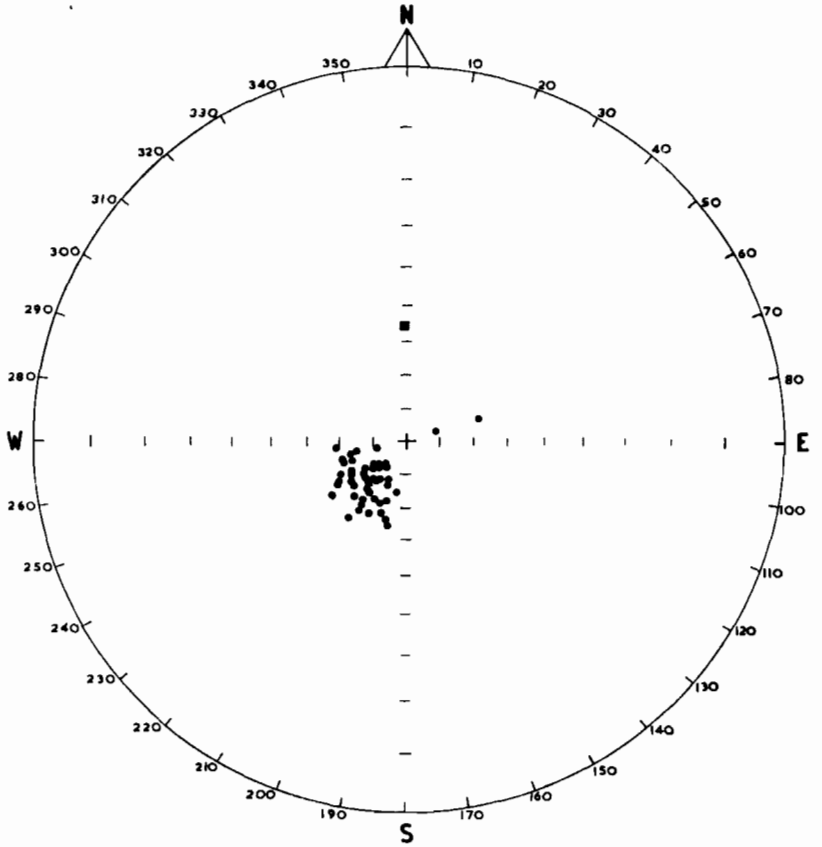


Figure 7A

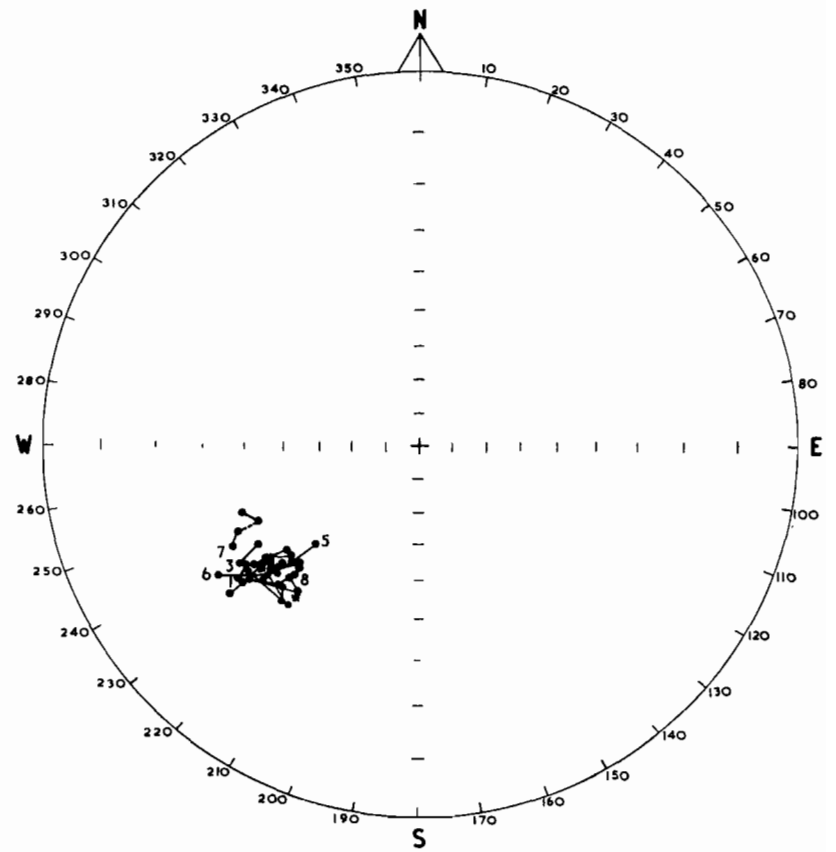


Figure 7B

Figure 7. Stereograms of samples from the south rim of the Grand Canyon, northern Arizona. A, Hakatai Shale; B, Bass Limestone

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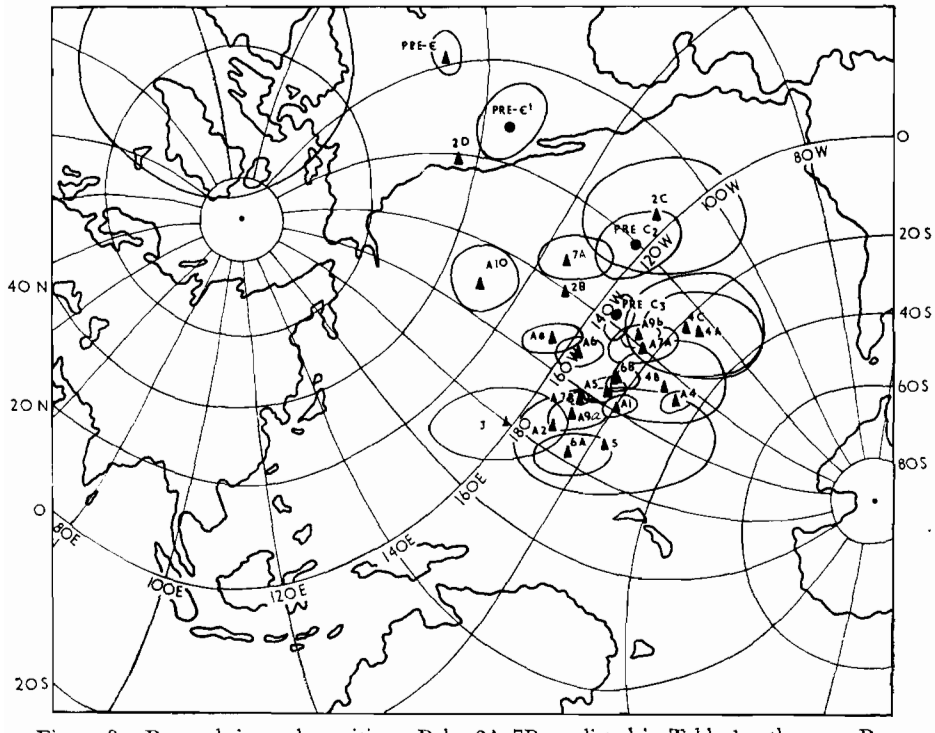


Figure 8. Precambrian pole positions. Poles 2A–7B are listed in Table 1; others are Precambrian poles from Figure 26 of Collinson and Runcorn (1960). (A1–A9b listed in Table 1 of this paper.)

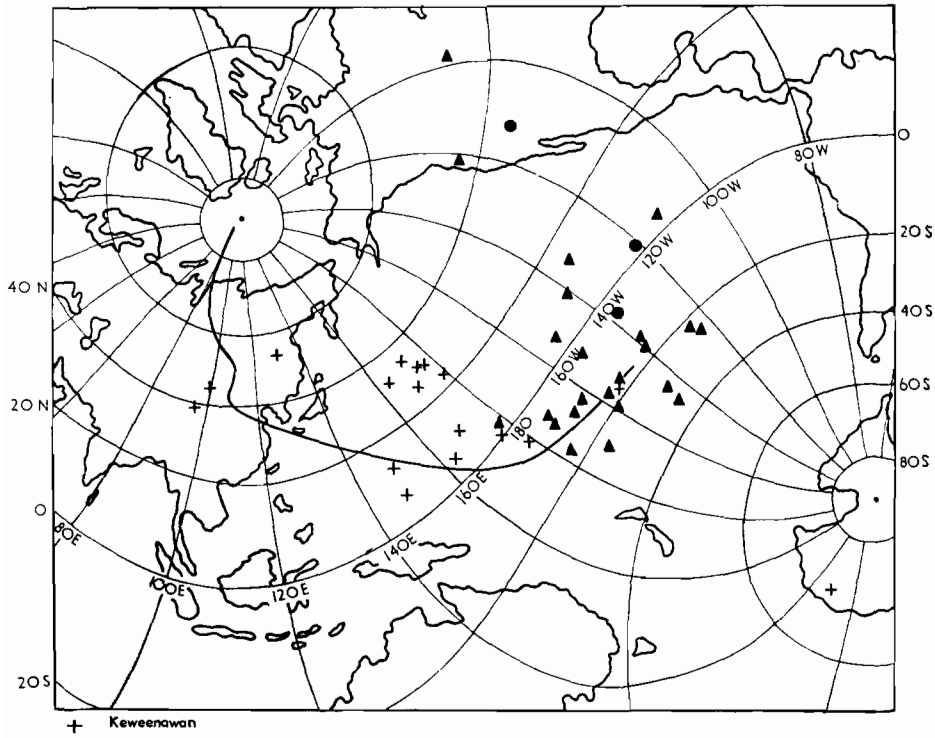


Figure 9. Keweenawan pole positions (DuBois, 1962) compared with those in this paper.

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of the sedimentary rocks reported in this paper are weak, these methods have not yet greatly improved the accuracy of the results and will not be reported until a later date.

The best proof that the magnetizations of these formations lie along the direction of the earth's magnetic field at or soon after their depositions is the good grouping of the pole positions calculated from the mean paleomagnetic directions *after* correction for geological dip. The large values of dip of the formations at the sites sampled are shown in the eighth column of Table 1. As the strikes are different, the pole positions would be greatly scattered had they been computed from the directions of magnetizations of the formations in their present field positions.

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